Validation and AI Use

Named record | current model plus separately qualified historical Lab11 evidence

Decision, version, and as-of boundary

WATCH/DEFER for a no-position buy-side committee. September16,2026 information cutoff; public sources retrieved October19. Five rolling annual periods, USDm/455m diluted proxy. Environment Python3.11.4; no third-party packages/network for valuation. Artifact-generation tools are separate.

Model SHA-256: 4705f796ac2e59722169fcfcfb41d74faf9dd76d2669b585f767cf05b978fa04

Run and input hashes:02_repository/outputs/run_manifest.json. Current technical cold-run log is linked in07_frozen_records/COLD_RUN_LOG.md; it is an actual fresh no-pip environment, but not a student-recorded video.

Data and convention ledger

Material input	Source / as-of availability	Definition / units	Transformation / exclusion	Risk
Original annual history	K25,FY2025, filed2026-02-19	Revenue11,100; EBIT2,277; CFO2,009 USDm	Normalized original EBIT+91+61=2,429; not silently mixed with recast basis	Recurring charges and tax cash effects
June balance	Q26,June30, availableAug5	Cash2,310; debt6,735; investments3,314 USDm	Exclude customer funds; split773 equity holdings from debt investments	Not September spot balance
Software basis	Q26 Note1; effective2026	December PPE1,338->1,165; equity4,615->4,483	Reconcile173 asset,41 liability and132 equity change	Original numbers correct on original basis
Operating scale	Q26,H1 endedJune30	Revenue6,223; GMV44,595 USDm	Double H1 for starting annual scale; separate ads/GMV and marketplace/GMV	Seasonality; not actual annual/LTM
Depop transaction	Q26 Note3; ET26 Note2; closeJuly30	Cash paid1,400 USDm; seller June assets/cash	Infer goodwill from book net assets; forecast operating business, no DCF cost addback	Closing cash/PPA/recurring burn unknown
Shares	Q26 coverJuly31 / Q2 Note2	445m common,10m awards	455m diluted proxy; not468 annual EPS shares	Spot dilution unverified
Dated price	PRICE_EB,September16 close	109.17 USD/share	Regular Close, not adjusted return price	Not latest October market price
Discount inputs	RF Sept16; judgments disclosed	Treasury5.01%; beta1; ERP5%; debt6.01%	Market equity/book debt weights;5% accounting interest separate	Beta/spread/market debt not established
Terminal	Forecast judgment	g2.5%, incremental ROIC15%	NOPAT*(1-g/ROIC); no free growth	Sustainability and70.59% terminal dependence
Investments/claims	Q26 June; analyst judgments	2,541 liquid;773 equity; cash floor1,000	Equity haircut15%; other claims0 assumed; cash shortfall deducted	Tax, liquidity, claims omissions
Peers	ET26 H1 annualized,September16 market	Continuing earnings; common-equity P/E and EV/EBIT	Depop proceeds once; current finance lease6.4m proxy; one qualified peer	Scope/capitalization/DST/convertibles/bridge
Authorship / lineage	Source ledger and review record	Facts/normalizations/forecasts/outputs separated	Student source checks reported; raw source snapshots/hashes retained	No grade or AI agreement as proof

Validation register

Claim / output	Failure mode	Test / expected	Actual	Disposition	Evidence
Filing and accounting-basis reconciliation	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	{"FY2025 reported income": 0, "FY2025 CFO reconciliation": 0, "FY2025 cash definition reconciliation": 0, "FY2025 full cash flow tie": 0, "FY2025 reported balance": 0, "2025 recast PP&E": 0, "2025 recast assets": 0, "2025 recast liabilities": 0, "2025 recast equity": 0, "June assets": 0, "June liabilities": 0, "June...	PASS; limits disclosed	outputs/validation.json
Five-year statements articulate	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	true	PASS; limits disclosed	outputs/validation.json
Synthetic closed-form known answer using actual DCF kernel	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	{"expected": 1000.0, "actual": 999.9999999999997, "units": "synthetic currency units", "limitation": "Independent synthetic identity; student reports no separate lab answer key was provided."}	PASS; limits disclosed	outputs/validation.json
Legacy training answer reproduced through final discount kernel	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	{"expected_from_lab_notes": 27.4974, "actual": 27.49737039819684, "status": "Reproduces lab-note benchmark; no separate instructor key available according to student. Grades do not prove every calculation."}	PASS; limits disclosed	outputs/validation.json
WACC/g monotonicity with positive terminal economics	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	{"wacc_lower_price": 63.379689902889204, "base_price": 53.16594423347071, "wacc_higher_price": 45.53511609032838, "g_low": 50.30333446236393, "g_high": 56.92393929896758}	PASS; limits disclosed	outputs/validation.json
Reject WACC <= terminal growth	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	Invalid WACC/growth/ROIC boundary	PASS; limits disclosed	outputs/validation.json
Reject terminal ROIC <= growth	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	Invalid WACC/growth/ROIC boundary	PASS; limits disclosed	outputs/validation.json
Highest-risk: typed cash must stop valuation	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	2027 asset sum: actual=19201.082571075043, expected=19301.082571075043, gap=-100.0	PASS; limits disclosed	outputs/validation.json
Broken CFO link refuses	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	2027 cash: actual=1053.7362185666477, expected=1063.7362185666477, gap=-10.0	PASS; limits disclosed	outputs/validation.json
NaN input refuses	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	inputs.assumptions.beta: invalid finite numeric input	PASS; limits disclosed	outputs/validation.json

Claim / output	Failure mode	Test / expected	Actual	Disposition	Evidence
Liquidity exhaustion refuses	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	2027 liquidity failure: revolver exhausted	PASS; limits disclosed	outputs/validation.json
FCFF excludes financing; no double debt subtraction	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	Distributions/paydown alter cash/debt, not operating FCFF or opening EV bridge.	PASS; limits disclosed	outputs/validation.json
Negative FCFF retained; no invented recovery	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	{"signed_present_value": -10989.543102581994, "incorrect_clipped_value": 0.0, "terminal_rejection": "Nonpositive terminal operating profit: explicit recovery required"}	PASS; limits disclosed	outputs/validation.json
Peer/multiple definition arithmetic	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	Common-equity numerator; continuing H1 denominators; debt not added to P/E.	PASS; limits disclosed	outputs/validation.json
Reverse DCF reprices target	Definition, articulation or boundary failure	Expected identity/direction or refusal; test code retained	{"status": "solved", "constant_annual_gmv_growth": 0.2007587312720716, "check_price": 109.16999999778305, "target": 109.17, "held_fixed": "Base ad/market yields, costs, Depop, WACC, terminal growth/ROIC, bridge and shares."}	PASS; limits disclosed	outputs/validation.json

Known answer uses the SAME final discount kernel:27.4973704 versus local lab27.4974, within0.0001;100-per-year perpetuity at10% gives1,000. Student reports no separate published lab key and full credit; the key/provenance discrepancy remains disclosed. Failed cases are visible as expected refusals; passing these refusals does not make forecast evidence complete.

Locked Changed-Input Record - human prediction first; AI closed during execution

Precommit before the run and before AI assistance on the prediction

Precommit timestamp	Repository commit	Material input	Old -> new / units	Expected direction	Expected decision effect
Notes:Sep29,2026, 5PM; timezone unrecorded	Not supplied; student reports due-date GitHub/Brightspace submission	Annual REVENUE growth, historical FY2026-2030 FCFE model	4% ->6%, +2pp each year; other independent inputs fixed	FY2030 EBIT+8%, FCFE+12%,share value+10%	NOT explicit in original pre-run text; do not fabricate retrospectively

Student explicitly confirms AI-off prediction authorship and saving before execution. Independent pre-run version artifact is not inspected. The newly supplied template ALSO requires AI closed during execution; that execution condition remains unconfirmed. This is a qualified historical record, not a claim that all locked-record requirements pass.

Preserved execution result

Before / after	Actual decision effect	Prediction reconciliation	Diagnosis / no change	Evidence
Approx60.24 ->64.48; EBIT +279.20m(10.08%);FCFE+183.43m(10.70%);share+4.24 (7.03%)	No reversal versus saved107.83, but dates differ	All directions matched; magnitudes differed	Compounding, reinvestment and multi-year valuation explain gap; expected decision effect absent	lab11_notes1.md; lab11_sensitivity_output.md/details.json; original unmodified

Historical model is December31,2025 FCFE/449m common shares, not current September16 FCFF/455m proxy. Restoration appears in saved lab output. New AI GMV check53.17->58.37->53.17 and ROIC-only checks are explicitly AI-assisted, never substituted as the human prediction.

Named AI-use register

Tool / exposed version	Interaction date	Material task	Output considered	Independent check	Disposition / decision effect
Codex workspace assistant; exact deployed model/version not retained	Oct18-19,2026	Organization and source research	Folder/template structure and source claims	Student reports source comparisons; public filing/price/Treasury records retained	Modify/qualify; keep actual evidence gaps
Codex; exact version unrecorded	Oct19	Financial engine implementation	Linked statements/FCFF-WACC and scenarios	Filing sums; shared-kernel known answer; negative flows; bridge; fresh reproduction	Use provisionally; judgment not delegated
Codex; exact version unrecorded	Oct19	Source/definition corrections	Original/recast PPE; annual EPS shares vs spot proxy; advertising basis	Q26 Note1/Note2, cover;K25 MD&A	Correct/qualify; prevent mixed-basis confidence
Codex; exact version unrecorded	Oct19	Sensitivity and AI-output challenge	GMV/ROIC tests; old positive-only flow rule	Signed deficit identity, independent source definitions and arithmetic	Reject clipped deficits; retain tentative range
Codex; exact version unrecorded	Oct19	Memo, video and PDF communication	AI-assisted drafts and consistent evidence map	Against saved numeric result/input hashes and supplied templates	Modify; preserve student ownership and missing human evidence

Limitations, monitoring, and kill/escalation rules

Financial: synthetic September cash/share/PPA, judgmental beta/spread/cost/terminal ROIC, simplified tax/SBC/working capital and single-peer scope remain provisional. Reporting accuracy and accounting identities do not establish economic forecasts.

Monitoring: assigned analyst reviews each release and filing within five business days. In EACH of two consecutive quarters, nonpositive FX-neutral GMV AND year-over-year ad-yield decline prompts re-underwriting. Funding shortfall after a separate distribution-cut assessment without recovery/committed financing prompts DO NOT INITIATE. Code does not automate that distribution cut. Entry80% of updated low value also needs verified bridge and fundable downside. Later outcomes are future evidence only.

Human/course: original baseline PDF and independently inspected receipt/commit history absent; original expected decision effect absent; AI-closed execution unconfirmed; no new partner approval; videos/corrected transcripts/access proof remain to be produced. No separate key was available according to student. Do not manufacture compliance from grades.

Source index

K25: eBay FY2025 Form 10-K (original basis) | published/observation 2026-02-19 | retrieved 2026-10-19 | https://www.sec.gov/Archives/edgar/data/1065088/000106508826000027/ebay-20251231.htm

Q26: eBay Q2 2026 Form 10-Q | published/observation 2026-08-05 | retrieved 2026-10-19 | https://www.sec.gov/Archives/edgar/data/1065088/000106508826000177/ebay-20260630.htm

ER26: eBay Q2 2026 earnings release | published/observation 2026-08-05 | retrieved 2026-10-19 | https://www.ebayinc.com/stories/news/ebay-inc-reports-second-quarter-2026-results/

ET26: Etsy Q2 2026 Form 10-Q | published/observation 2026-08-05 | retrieved 2026-10-19 | https://investors.etsy.com/sec-filings/all-sec-filings/content/0001370637-26-000080/etsy-20260630.htm

PRICE_EB: EBAY price history, September 16 Close | published/observation 2026-09-16 | retrieved 2026-10-19 | https://www.quiverquant.com/stock/EBAY/historical-prices/

PRICE_ET: ETSY price history, September 16 Close | published/observation 2026-09-16 | retrieved 2026-10-19 | https://stockanalysis.com/stocks/etsy/history/

RF: Treasury daily par yield curve, September 16, 10-year | published/observation 2026-09-16 | retrieved 2026-10-19 | https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?field_tdr_date_value=2026&type=daily_treasury_yield_curve

RF2: FRED DGS10, September 16 | published/observation 2026-09-16 | retrieved 2026-10-19 | https://fred.stlouisfed.org/graph/?g=1pNoR

Exact fact values, source table/page locators and transformations:02_repository/data/finance_inputs.json and outputs/VISIBLE_RESULTS.md. Proof paths/commands/failures:validation.json, COLD_RUN_LOG.md and linked clean-run JSON. Original lab and Edition A files remain unchanged.